

Sprint 1

Epic 1: Data Collection

- Collect historical rainfall dataset (USN-1) – 3
- Collect river water level dataset (USN-2) – 2
- Collect weather and flood history data (USN-3) – 3

Epic 2: Visualizing and Analysing the Data

- Generate rainfall graphs (USN-4) – 2
- Visualize flood trends using charts (USN-5) – 3

Total Story Points in Sprint 1

$$3 + 2 + 3 + 2 + 3 = 13$$

Sprint 2

Epic 3: Data Pre-processing

- Handle missing values (USN-6) – 3
- Remove duplicate records (USN-7) – 2
- Normalize and clean dataset (USN-8) – 3

Epic 4: Model Building

- Train XGBoost flood prediction model (USN-9) – 5
- Evaluate model performance (USN-10) – 3

Total Story Points in Sprint 2

$$3 + 2 + 3 + 5 + 3 = 16$$

Sprint 3

Epic 5: Application Building

- Develop Flask backend APIs (USN-11) – 5
- Design web interface (USN-12) – 3
- Integrate ML model with application (USN-13) – 5
- Display prediction results and alerts (USN-14) – 3
- Perform testing and deployment (USN-15) – 2

Total Story Points in Sprint 3

$$5 + 3 + 5 + 3 + 2 = 18$$

Total Story Points

Sprint	Story Points
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Sprint 1	13
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Sprint 2	16
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Sprint 3	18
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$$\text{Total Story Points} = 13 + 16 + 18 = 47$$

Velocity Calculation

Velocity = Total Story Points Completed ÷ Number of Sprints

Total Story Points = **47**

Number of Sprints = **3**

$$\text{Velocity} = 47 \div 3 = 15.67 \approx 16 \text{ Story Points per Sprint}$$

Final Result

- **Sprint 1 = 13 Story Points**
- **Sprint 2 = 16 Story Points**
- **Sprint 3 = 18 Story Points**

Total Story Points = 47

Team Velocity = 16 Story Points per Sprint

This sprint breakdown aligns with your project phases (**Data Collection → Visualization → Data Pre-processing → Model Building → Application Building**) and follows the same format as the example you shared.